

Problem 11: Bulk Purchase Discount

Domain: Retail **CE:** test **Built-in functions:** >= (comparison), * (arithmetic), - (arithmetic)

Background

A store applies a 10% discount to any purchase where the quantity meets or exceeds a minimum threshold. The discount is applied to the total price.

Given Data (provided in deffacts)

```
(deftemplate purchase (slot item) (slot price) (slot quantity))
(deffacts store-data
  (discount-threshold 3)
  (purchase (item Laptop) (price 15000) (quantity 1))
  (purchase (item Notebook) (price 50) (quantity 5)))
```

Task

Write a CLIPS program that identifies bulk purchases, computes the discounted total, and prints the result.

Requirements

- Use a deftemplate fact for purchase data and an ordered fact for the discount threshold
- Use a test CE to check if quantity meets the threshold
- Use >= to detect bulk purchases, * to compute totals, and - to subtract the discount
- Print the item name and discounted total

Expected Output

```
Notebook discounted total: 225.0 EGP
```

Problem 12: Account Login Lockout

Domain: Cybersecurity **CE:** not **Built-in functions:** >= (comparison), integerp (predicate)

Background

A system locks accounts after a set number of failed login attempts. Lockout is only applied if no clearance has been previously granted for that user.

Given Data (provided in deffacts)

```
(deftemplate user (slot username) (slot failed-attempts))
(deffacts system-data
  (lockout-threshold 3)
  (user (username admin) (failed-attempts 4))
  (user (username guest) (failed-attempts 1)))
```

Task

Write a CLIPS program that locks out users who have exceeded the failed attempt threshold and have no existing clearance.

Requirements

- Use a deftemplate fact for user data and an ordered fact for the threshold
- Use a not CE to check that no clearance fact exists for the user
- Use >= to detect exceeded attempts, and integerp to verify the attempt count is an integer
- Print the username and failed attempt count

Expected Output

```
LOCKED: admin - 4 failed attempts
```

Problem 13: Daily Calorie Alert

Domain: Nutrition **CE:** exists **Built-in functions:** > (comparison), - (arithmetic)

Background

A nutrition tracking system alerts the user if any single meal exceeds the per-meal calorie limit, then lists all meals that exceed it and by how much.

Given Data (provided in deffacts)

```
(deftemplate meal (slot name) (slot calories))
(deffacts nutrition-data
  (daily-log)
  (meal-limit 600)
  (meal (name Breakfast) (calories 450))
  (meal (name Lunch) (calories 750))
  (meal (name Dinner) (calories 500)))
```

Task

Write a CLIPS program that first alerts if any meal exceeds the limit, then prints each excessive meal with its excess amount.

Requirements

- Use a deftemplate fact for meal data and ordered facts for the trigger and limit
- Use an exists CE to check if at least one meal exceeds the calorie limit
- Use > to compare calories to the limit and - to compute the excess
- Print a general alert from exists, then list each excessive meal

Expected Output

```
ALERT: At least one meal exceeds the 600 calorie limit
Lunch exceeds limit by 150 cal
```

Problem 14: Dean's List Eligibility

Domain: Education **CE:** test **Built-in functions:** >= (comparison), and (logical), floatp (predicate)

Background

A student is placed on the Dean's List if their GPA is 3.7 or above AND they have completed at least 100 credit hours. Both conditions must be satisfied simultaneously.

Given Data (provided in deffacts)

```
(deftemplate student (slot name) (slot gpa) (slot credits))
(deffacts registrar-data
  (honors-review)
  (student (name Ahmed) (gpa 3.8) (credits 120))
  (student (name Layla) (gpa 3.5) (credits 90))
  (student (name Omar) (gpa 4.0) (credits 105)))
```

Task

Write a CLIPS program that identifies eligible students and prints those on the Dean's List.

Requirements

- Use a deftemplate fact for student data and an ordered fact as a trigger
- Use a test CE with and to combine both conditions
- Use >= for both threshold comparisons, floatp to verify the GPA type
- Print the name and GPA of each eligible student

Expected Output

```
Ahmed is on the Dean's List - GPA: 3.8
Omar is on the Dean's List - GPA: 4.0
```

Problem 15: Planet Zone Classifier

Domain: Astronomy **CE:** or **Built-in functions:** < (comparison), > (comparison), float (conversion)

Background

Planets are classified as being in an 'extreme zone' if they are closer than 2.0 AU (inner zone, high radiation) or farther than 5.0 AU (outer zone, extreme cold) from the Sun.

Given Data (provided in deffacts)

```
(deftemplate planet (slot name) (slot distance-au))
(deffacts solar-system
  (zone-check)
  (planet (name Mercury) (distance-au 0.39))
  (planet (name Earth) (distance-au 1.0))
  (planet (name Saturn) (distance-au 9.58)))
```

Task

Write a CLIPS program that classifies planets in extreme zones and prints their name and distance.

Requirements

- Use a deftemplate fact for planet data and an ordered fact as a trigger
- Use an or CE combining two test conditions
- Use < and > for zone comparisons, and float to display the distance
- Print each planet in an extreme zone along with its distance in AU

Expected Output

```
Mercury at 0.39 AU - extreme zone
Saturn at 9.58 AU - extreme zone
```

Problem 16: Vehicle Maintenance Scheduler

Domain: Fleet Management **CE:** or + and **Built-in functions:** - (arithmetic), > (comparison), >= (comparison)

Background

A vehicle needs servicing if EITHER: (a) it has exceeded the oil change interval AND has high mileage, OR (b) its tires are old AND mileage is above a threshold.

Given Data (provided in deffacts)

```
(deftemplate vehicle (slot plate) (slot mileage) (slot oil-change-km) (slot tire-age))
(deffacts fleet-data
  (maintenance-check)
  (service-interval 10000)
  (vehicle (plate V001) (mileage 85000) (oil-change-km 72000) (tire-age 4))
  (vehicle (plate V002) (mileage 60000) (oil-change-km 58000) (tire-age 1)))
```

Task

Write a CLIPS program that flags vehicles needing service based on either oil change overdue or tire age conditions.

Requirements

- Use a deftemplate fact for vehicle data and ordered facts for the trigger and interval
- Use an or CE with nested and CEs for the two service conditions
- Use - to compute km since oil change, > and >= for threshold comparisons
- Print the plate and mileage of vehicles needing service

Expected Output

```
Service required: V001 at 85000 km
```

Problem 17: Heart Rate Monitor

Domain: Healthcare **CE:** or **Built-in functions:** > (comparison), < (comparison), integerp (predicate)

Background

A cardiac monitoring system flags patients with abnormal heart rates — either above 100 bpm (tachycardia) or below 50 bpm (bradycardia).

Given Data (provided in deffacts)

```
(deftemplate patient (slot name) (slot heart-rate) (slot age))
(deffacts cardiac-data
  (monitor active)
  (patient (name Salma) (heart-rate 115) (age 45))
  (patient (name Tamer) (heart-rate 40) (age 30))
  (patient (name Amal) (heart-rate 72) (age 55)))
```

Task

Write a CLIPS program that detects patients with abnormal heart rates and prints a cardiology alert for each.

Requirements

- Use a deftemplate fact for patient data and an ordered fact as a trigger
- Use an or CE combining two threshold tests
- Use > and < for threshold comparisons, and integerp to verify heart-rate type
- Print the patient name and heart rate

Expected Output

```
ALERT: Salma has abnormal heart rate: 115 bpm
ALERT: Tamer has abnormal heart rate: 40 bpm
```

Problem 18: Air Pollution Alert

Domain: Environment **CE:** not **Built-in functions:** > (comparison), abs (math), - (arithmetic)

Background

A pollution monitoring system alerts authorities when the Air Quality Index (AQI) exceeds 100. Alerts are only issued when no remediation plan is already in place.

Given Data (provided in deffacts)

```
(deftemplate sensor (slot location) (slot aqi))
(deffacts environment-data
  (monitoring active)
  (sensor (location Downtown) (aqi 180))
  (sensor (location Suburbs) (aqi 45)))
```

Task

Write a CLIPS program that flags high-pollution areas and prints an alert with the deviation from safe levels, only if no remediation plan is active.

Requirements

- Use a deftemplate fact for sensor data and an ordered fact as a trigger
- Use a not CE to suppress alerts when a remediation plan exists
- Use > to detect high AQI, - to compute deviation, and abs to ensure a positive value
- Print the location, AQI level, and deviation from the safe threshold

Expected Output

```
ALERT: Downtown AQI=180 - deviation: 80
```


Problem 19: Recipe Ingredient Scaler

Domain: Cooking **CE:** test **Built-in functions:** * (arithmetic), > (comparison), float (conversion)

Background

A recipe scaling tool computes total ingredient amounts needed for a given number of servings. Items requiring more than 100 units total should be flagged for bulk purchase.

Given Data (provided in deffacts)

```
(deftemplate ingredient (slot name) (slot amount-per-serving))
(deffacts recipe-data
  (servings-needed 6)
  (ingredient (name Flour) (amount-per-serving 50))
  (ingredient (name Eggs) (amount-per-serving 1))
  (ingredient (name Butter) (amount-per-serving 30)))
```

Task

Write a CLIPS program that computes the total quantity for each ingredient and flags those needing bulk purchase.

Requirements

- Use a deftemplate fact for ingredient data and an ordered fact for servings needed
- Use a test CE to check if total exceeds 100 units
- Use * to scale amounts and > to compare, and float to display the total
- Print each ingredient requiring bulk purchase and its total amount

Expected Output

```
Flour: 300.0 units (buy in bulk)
Butter: 180.0 units (buy in bulk)
```

Problem 20: Low Balance Alert

Domain: Banking **CE:** not **Built-in functions:** < (comparison), numberp (predicate)

Background

A bank sends low-balance alerts to customers whose balance falls below 100 EGP. The alert is only sent if a notification has not already been sent to that customer.

Given Data (provided in deffacts)

```
(deftemplate account (slot owner) (slot balance))
(deffacts bank-data
  (daily-check)
  (account (owner Ahmed) (balance 50))
  (account (owner Sara) (balance 2000)))
```

Task

Write a CLIPS program that flags low-balance accounts and sends an alert only if no notification has been sent previously.

Requirements

- Use a deftemplate fact for account data and an ordered fact as a trigger
- Use a not CE to suppress alerts if a notification was already sent
- Use < to detect low balance and numberp to verify the balance is numeric
- Print the owner name and their balance

Expected Output

```
Low balance alert for Ahmed: 50 EGP
```

Problem 21: Meeting Conflict Detector

Domain: Scheduling **CE:** exists **Built-in functions:** < (comparison), > (comparison), - (arithmetic)

Background

Two meetings conflict if one starts before the other ends and ends after the other starts. The system should flag any meeting that overlaps with at least one other meeting.

Given Data (provided in deffacts)

```
(deftemplate meeting (slot title) (slot start) (slot end))
(deffacts schedule-data
  (conflict-check)
  (meeting (title Standup) (start 9) (end 10))
  (meeting (title ClientCall) (start 9) (end 11))
  (meeting (title Lunch) (start 12) (end 13)))
```

Task

Write a CLIPS program that flags conflicting meetings and prints the duration of all meetings.

Requirements

- Use a deftemplate fact for meeting data and an ordered fact as a trigger
- Use an exists CE to check if any other meeting overlaps
- Use < and > for overlap detection, and - to compute each meeting's duration
- Print flagged meetings from exists, and print duration for all meetings

Expected Output

```
CONFLICT: Standup overlaps with another meeting
CONFLICT: ClientCall overlaps with another meeting
Standup: 1 hour(s) duration
ClientCall: 2 hour(s) duration
Lunch: 1 hour(s) duration
```

Problem 22: Velocity Limit Check

Domain: Physics **CE:** test **Built-in functions:** / (arithmetic), > (comparison), float (conversion)

Background

Velocity is computed from distance and time. Objects exceeding the speed limit must be flagged for investigation.

Given Data (provided in deffacts)

```
(deftemplate motion (slot object) (slot distance) (slot time-sec))
(deffacts physics-data
  (speed-limit 30.0)
  (motion (object Car-A) (distance 600) (time-sec 15))
  (motion (object Car-B) (distance 100) (time-sec 10)))
```

Task

Write a CLIPS program that computes velocity for each object and prints those exceeding the speed limit.

Requirements

- Use a deftemplate fact for motion data and an ordered fact for the speed limit
- Use a test CE to filter objects exceeding the speed limit
- Use / to compute velocity and > to compare against the limit, and float to display
- Print the object name and its velocity

Expected Output

```
Car-A exceeds limit: 40.0 m/s
```

Problem 23: Building Fire Risk Assessment

Domain: Emergency Management **CE:** or **Built-in functions:** > (comparison), * (arithmetic)

Background

A building is classified as high fire risk if it has more than 20 floors OR if the product of its floors and occupants exceeds 5000.

Given Data (provided in deffacts)

```
(deftemplate building (slot name) (slot floors) (slot occupants))
(deffacts emergency-data
  (inspection-active)
  (building (name TowerA) (floors 25) (occupants 500))
  (building (name StoreB) (floors 2) (occupants 15)))
```

Task

Write a CLIPS program that identifies high-risk buildings and prints an inspection alert for each.

Requirements

- Use a deftemplate fact for building data and an ordered fact as a trigger
- Use an or CE combining two test conditions
- Use > to compare floors, and * and > for the occupant-density check
- Print the building name and an alert message

Expected Output

```
HIGH FIRE RISK: TowerA - immediate inspection required
```

Problem 24: Harvest Readiness Check

Domain: Horticulture **CE:** test **Built-in functions:** >= (comparison), - (arithmetic)

Background

A crop is ready to harvest when the number of days since planting meets or exceeds the required days to harvest.

Given Data (provided in deffacts)

```
(deftemplate crop (slot name) (slot planted-day) (slot current-day) (slot days-to-harvest))
(deffacts farm-data
  (harvest-check)
  (crop (name Wheat) (planted-day 1) (current-day 95) (days-to-harvest 90))
  (crop (name Rice) (planted-day 10) (current-day 90) (days-to-harvest 120)))
```

Task

Write a CLIPS program that checks if crops are ready for harvest and prints the result.

Requirements

- Use a deftemplate fact for crop data and an ordered fact as a trigger
- Use a test CE to check if days grown meets the harvest threshold
- Use - to compute days grown and >= to compare against days-to-harvest
- Print the crop name and total days grown

Expected Output

```
Wheat is ready to harvest! Days grown: 94
```